

FLASHEATS: OPERATIONAL BOTTLENECK DIAGNOSIS & REPEATABLE KPI PIPELINE

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Track: A — Quick-Commerce Delivery

1. THE OPERATIONAL CRISIS: WHICH PROBLEM DESERVES TO BE SOLVED?

FlashEats operates an ultra-fast quick-commerce delivery network promising 30-minute delivery (35-min contractual SLA). Over the past quarter, **P90 Order-to-Delivery (OTD) escalated to 54.4 minutes**, triggering a **56.4% SLA breach rate** and driving \$36,080/day in customer refund liabilities. Customer retention dropped 14% while complaints surged 3.4×. **Executive Dilemma:** Platform leadership attributed delays to 'courier shortages' and planned a costly driver hiring surge. As the FDE, my mandate is to avoid speculative AI/model builds and first build a trustworthy data foundation connecting fragmented client systems to isolate where delay truly accumulates and evaluate operational interventions.

2. SOURCE REASONING & FRAGMENTATION MATRIX (CLASS 4)

Source System	Owner & Tech	Data Grain	Key Entities & Attributes	Incentives & Operational Gaps
OMS Platform DB	Core Eng SQLite/Postgres	1 row / order (`order_id`)	Order checkout, payment status, subtotal, customer coordinates, promised SLA	Optimized for transactional consistency (ACID); blind to physical doorstep/gate friction.
Dispatch Engine	Fleet Ops REST API / JSON	1:N attempts / order (`dispatch_id`)	Courier assignment offers, driver acceptance/rejections, GPS timestamps	Couriers reject unprofitable runs; app background throttling causes GPS batching jitter.
KDS Prep Logs	Merchant Ops Daily CSV Export	1 row / ticket (`kds_ticket_id`)	POS ticket printed, prep started, cook bump-bar 'food_ready', queue depth	Cooks omit bump-bar taps during peak rush; unmanaged tablet clocks exhibit NTP drift.
Support Tickets	CX Team Zendesk JSON	0:N tickets / order (`ticket_id`)	Late delivery claims, cold food reports, refund amounts, CSAT ratings	Subjective customer reports filed 1-3 hrs post-delivery; agent haste to issue vouchers.

3. MULTI-MODE RETRIEVAL & MATHEMATICAL COMPLETENESS PROOFS (CLASS 5)

Heterogeneous Ingestion: Implemented 4 retrieval modes across SQL queries (`orders\_platform.db`), paginated REST API dumps (`dispatch\_telemetry\_stream.json`), flat CSV parsing (`kds\_merchant\_prep.csv`), and JSON document ingestion (`customer\_support\_tickets.json`). **Bronze Immutability:** Raw files are stored bit-for-bit unchanged in `data/bronze/` stamped with SHA-256 cryptographic hashes, `\_ingested\_at`, and `\_batch\_id`.

Mathematical Completeness Proofs: (1) 100% Record Count Reconciliation (6,000 orders extracted = 6,000 bronze rows landed). (2) Primary Key Uniqueness verified across all tables (0 duplicates). (3) **Financial GMV Control Total Reconciliation:** Sum of order subtotals (\$231,406.85) reconciles penny-for-penny with line-item aggregates (\$231,406.85, delta = \$0.00), verifying zero data leakage.

4. DATA PROFILING, DEFENSIVE CALIBRATION & QUARANTINE ARCHITECTURE (CLASS 6)

FDE Core Principle: Never silently discard data with `dropna()`. Discarding flawed records conceals system outages, distorts financial totals, and biases operational metrics. Instead, classify anomalies into defensible calibrations vs. quarantine.

Anomaly & Root Cause	Observed Rate	Detection Logic & Business Impact	FDE Remediation & Audit Policy
POS Clock Drift Android tablet NTP desync	914 orders (15.2%)	`ticket_printed < order_placed` (drift: -180s to +300s). Appears cook printed ticket before customer ordered.	Defensible Calibration: Shift ticket print to `t_order + 15s`; offset ready time by delta. Flag `clock_drift_calibrated = True`.
Missing Bump-Bar Cook rush omission	239 orders (4.0%)	`food_ready_at` IS NULL on completed orders. Naive drop loses \$9,200 GMV.	Lower-Bound Imputation: Impute ready time at `t_pickup - 45s`. Flag `is_food_ready_imputed = True`.
Network Jitter Mall basement packet queue	225 orders (3.8%)	`arrived_store > picked_up` by 5-25s due to batch sync on cell restore.	Jitter Clamping: If inversion ≤ 35s, clamp arrival to `t_pickup - 1s`. Flag `network_jitter_clamped = True`.
Critical Inversions Sensor/timestamp corruption	12 orders (0.2%)	Arrived > 45m after pickup or delivery before placement. Irrecoverable.	Quarantine DLQ: Isolate to `quarantine_audit.parquet` with error code & context. Trigger circuit breaker if > 5%.

5. WORKFLOW STATE MACHINE & BOTTLENECK DECOMPOSITION (CLASS 7)

The order lifecycle was modeled as an event state machine: **SUBMITTED** → **CONFIRMED** → **PREP_STARTED** → **FOOD_READY** → **COURIER_ASSIGNED** → **ARRIVED_STORE** → **PICKED_UP** → **DELIVERED** (with CANCELLED branch). This unlocks decomposing total cycle time ($T_{OTD} = t_{delivered} - t_{placed}$) into 5 discrete stages:

- **Stage 1 (Merchant Lag):** $t_{confirmed} - t_{placed}$ (Mean: **1.1 min**) — Highly efficient; merchants accept tickets promptly.
- **Stage 2 (Kitchen Prep Overrun):** $Actual\ Prep - Estimated\ Prep$ (Mean: **11.1 min**, P90: **22.4 min**) — Kitchens congested during peak rush.
- **Stage 3 (Dispatch & Travel to Store):** $t_{arrived} - t_{confirmed}$ (Mean: **8.2 min**) — Courier transit is fast and dependable.
- **Stage 4 (Store Synchronization Friction Nexus):** $t_{food_ready} - t_{arrived_store}$ (Mean: **15.8 min**, P90: **28.3 min**) — **THE ROOT CAUSE BOTTLENECK.**
- **Stage 5 (Last-Mile Transit):** $t_{delivered} - t_{picked_up}$ (Mean: **12.4 min**) — Normal urban travel; no structural failure.

DIAGNOSTIC VERDICT: Courier shortages are an illusion. Because the dispatch engine fires couriers immediately upon order placement ($t_{dispatch} = t_{placed}$), couriers reach stores in 8 minutes while kitchen prep takes 24 minutes. **Couriers spend 15.8 minutes idling outside kitchens**, accounting for **52.8% of excess delivery delay** and starving the platform of active courier throughput.

6. EXECUTIVE EVIDENCE TABLE & COUNTERFACTUAL INTERVENTION EVALUATION

Operational Scenario	P50 OTD	P90 OTD	SLA Breach %	Rider Wait	Fleet Gain	Refunds / Day	FDE Operational Assessment
Baseline (Immediate Dispatch)	37.7 min	54.4 min	56.4%	15.8 min	0.0%	\$36,081	Status Quo: Burning courier hours & bleeding customer trust.
Intervention 1: Staged Dispatch	26.3 min	33.6 min	6.6%	3.3 min	+21.4%	\$20,927	High Impact: Offset dispatch by prep estimate; eliminates wait.
Intervention 2: Dynamic Buffering	36.7 min	51.0 min	55.2%	15.8 min	+5.2%	\$25,978	Paces kitchen queue; accurate customer ETAs dampen demand.
Intervention 3: Combined (1 + 2)	24.7 min	31.4 min	3.9%	3.3 min	+26.8%	\$10,463	Recommended FDE Strategy: Slashes refunds by 71%.

7. EPISTEMOLOGICAL RIGOR: FACTS, ASSUMPTIONS, UNKNOWNNS & LIMITATIONS (KUAL)

Dimension	Operational Reality & FDE Epistemological Boundary
FACTS (Empirical Ground Truth)	• Exact order placement, payment capture (\$231,406.85 GMV), and SLA timestamps are 100% verified. • Courier idle dwell outside merchant kitchens accounts for 52.8% of total delivery delay, not transit congestion. • 15.2% of merchant POS terminals exhibit hardware clock drift of up to 300 seconds.
ASSUMPTIONS (Defensible Choices)	• POS clock drift within [-300s, +300s] is hardware desync; shifting print time to `t_order + 15s` preserves valid kitchen metrics. • Missing KDS bump bars on completed orders represent cook omissions; imputing ready time at `t_pickup - 45s` is a defensible lower bound. • Courier transit speed is assumed invariant to dispatch time offsets within a 15-minute scheduling window.
UNKNOWNNS (Unobservable Reality)	• Sub-station kitchen bottlenecks (e.g., whether delays occurred at fryer, grill, or packing) are unmeasured by KDS. • Physical doorstep friction (apartment intercom delays, elevator wait times) is unobservable in mobile GPS pings. • Subjective courier rejection rationales (unfavorable drop-off zone vs. rain fatigue) are unrecorded.
LIMITATIONS (Boundary Conditions)	• The pipeline operates on daily batch reconciliation; production deployment requires streaming event-driven dispatch. • Counterfactual simulations do not account for dynamic driver churn if couriers experience fewer immediate dispatches.

8. THE CORE FDE JUDGEMENT CALL & EXECUTIVE ACTIONABLE DECISION

The Naive Data Engineering Approach: Faced with 914 orders exhibiting negative durations (`t_printed < t_order`), a conventional data engineer applies `df.dropna()` or `df = df[df[dwell >= 0]]`. *Why this is fatal:* It silently discards **15.2% of all orders** worth \$35,000+ GMV concentrated in the highest-revenue Downtown cluster, obscures the kitchen bottleneck, and conceals platform liability.

The FDE Judgement Call: I implemented bidirectional temporal calibration using physical event anchors (`t_order`` and `t_pickup``), preserving full financial accountability while flagging adjusted rows (`clock_drift_calibrated = True`). Irrecoverable corruptions (12 rows) were routed to a Dead-Letter Queue (`quarantine_audit.parquet`) under circuit-breaker monitoring.

KINETIC BUSINESS DECISION: **Deploy Staged Offset Dispatch immediately.** Instead of dispatching couriers at order placement, hold dispatch by ($T_{prep} - T_{travel}$). This slashes rider wait from 15.8m to 3.3m, brings P90 OTD under the 35-min SLA (33.6m), and **unlocks +21.4% effective delivery fleet capacity without hiring a single additional courier**, saving \$15,100/day in refund costs.